

CHEM 101 Topic 3: Matter and Energy — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work and reasoning to receive full credit.

Section 1: States of Matter & Phase Changes

Problem 1: A substance is highly compressible and fills the entire volume of any container it is placed in. Which state of matter is this?

Problem 2: Dry ice (solid CO_2) turns directly into a gas at room temperature. What is the name of this phase change, and is energy absorbed or released?

Problem 3: Explain why a person with severe dehydration may have a harder time cooling their body through sweating.

Section 2: Classifying Matter

Problem 4: Is distilled water (H_2O) an element, a compound, or a mixture? Explain your choice.

Problem 5: A suspension of an antibiotic medication must be shaken before use because the particles settle at the bottom. Is this a homogeneous or heterogeneous mixture?

Problem 6: Which separation technique would be most appropriate to remove salt from sea water to produce pure liquid water?

Section 3: Physical vs. Chemical Changes

Problem 7: Identify the following as a physical or chemical change: a) Dissolving sugar in hot tea; b) Digesting a piece of bread; c) Folding a piece of filter paper; d) A silver coin tarnishing over time.

Problem 8: You mix two clear liquids in a test tube, and the mixture suddenly becomes hot to the touch. Is this a physical or chemical change? Justify your answer.

Section 4: Conservation of Mass and Energy

Problem 9: In a chemical reaction, 12.0 g of substance X reacts with 8.0 g of substance Y. If 4.0 g of gas escapes from the open beaker, what is the mass of the remaining solid product?

Problem 10: Describe how the Law of Conservation of Energy applies to a patient's metabolism (the conversion of food into body heat and movement).